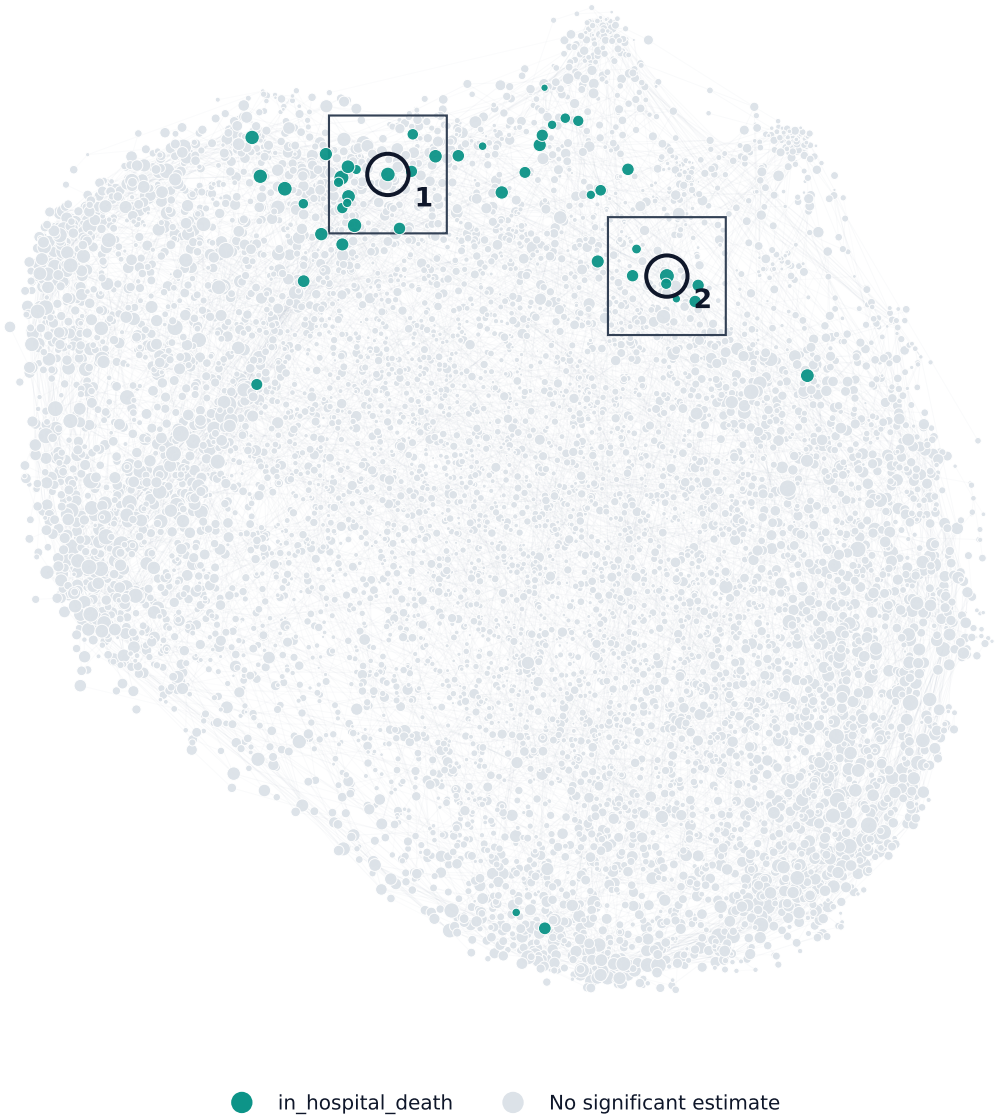
GLYPH / DISCOVERY BRIEF

Glyph + DINOv3: chest X-ray mortality

46 significant associations · 8192 concepts mapped

1/31 evaluable hits reproduce in test; 15 discovery hits lack evaluable comparison estimates.

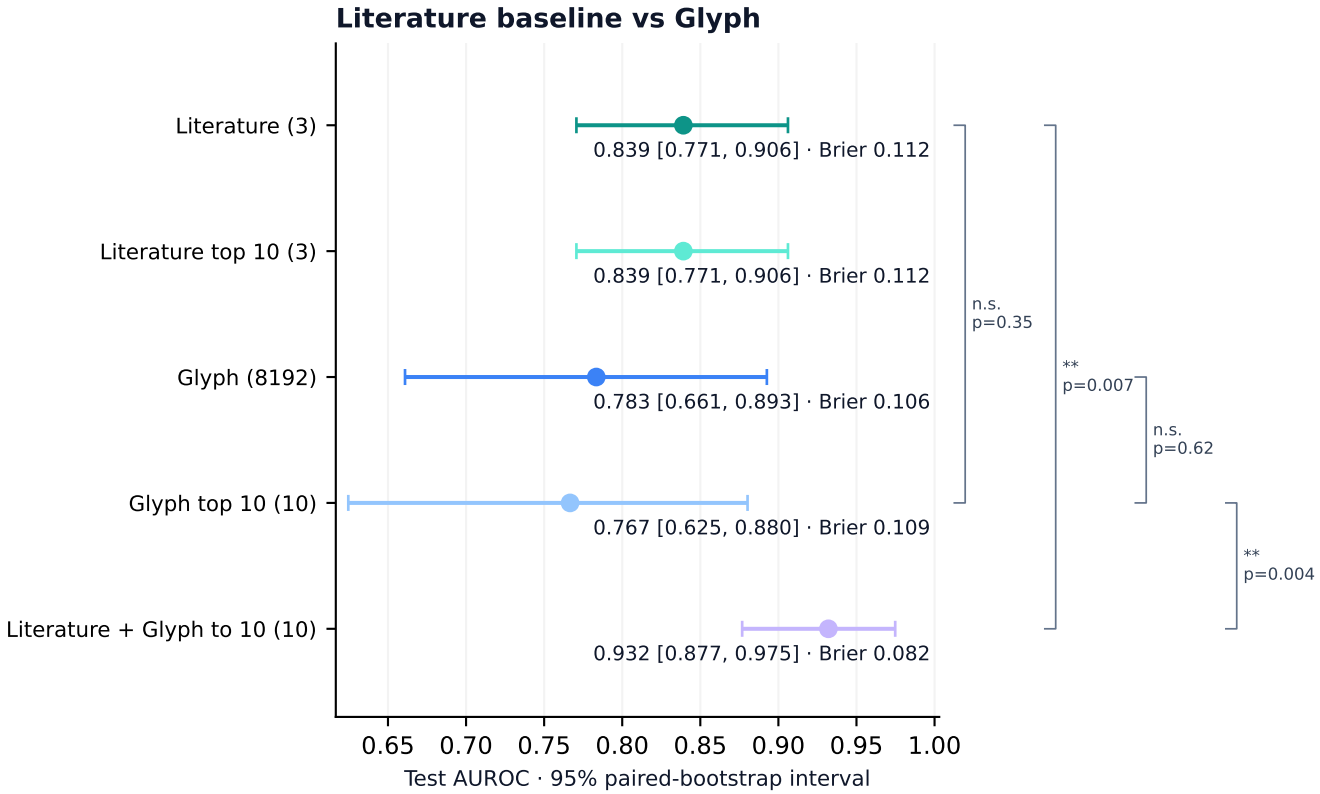
train 663 (115 events) / val 142 (24 events) / test 143 (25 events)
Reconstruction R^2: validation 0.905; test 0.902 · k=768
Test probe AUC: input vectors 0.768; full dictionary 0.781



TCIA COVID-19-NY-SBU (10.7937/TCIA.BBAG-2923); complete training-alive concept space. Color: strongest significant outcome. Node area is proportional to activation prevalence, with a tiny visibility floor and a common density adjustment. Missing prevalences use a uniform small marker. Lines show saved correlations, with at most eight per concept.

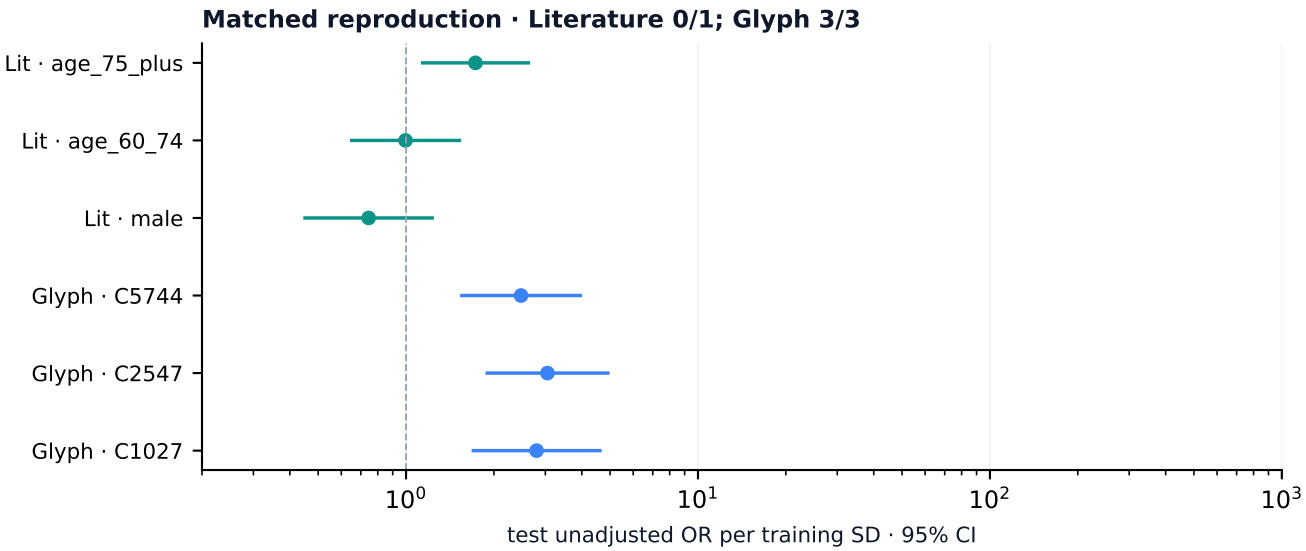
One patient per observation; internal validation at one site. Admission-period images precede the hospital-day-two mortality landmark. Age bands and sex form a simplified literature baseline, not the published model. No LLM interpretation or confounder adjustment. Interpretation not requested; cached interpretations are shown when available.

PREDICTION / BASELINE COMPARISON



Capped combination: ΔAUC +0.0929 (95% CI +0.0311 to +0.1619), paired p=0.00699 (exploratory, unadjusted).

Feature rankings use training only; validation selects ridge penalties and the best optimizer checkpoint. Combined models penalize literature and Glyph separately. Parentheses give predictor counts. The capped combined model retains up to ten training-ranked literature predictors and fills any remaining places with the highest-ranked Glyph concepts.

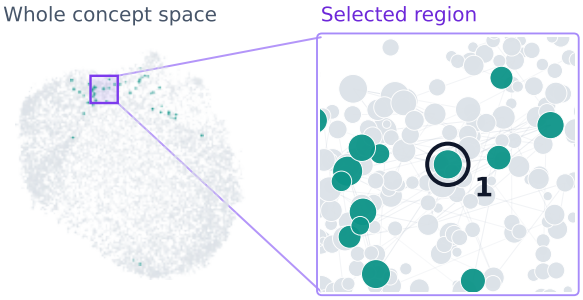


3 available literature markers; 8192 available Glyph concepts. Individual associations use 3 per source (6-test correction), separately from the full discovery family of 6878. Brackets show paired-bootstrap p-values, unadjusted (*<.05, **<.01, ***<.001; n.s. otherwise). Bootstrap intervals condition on fitted models. These results do not establish novelty or external validation.

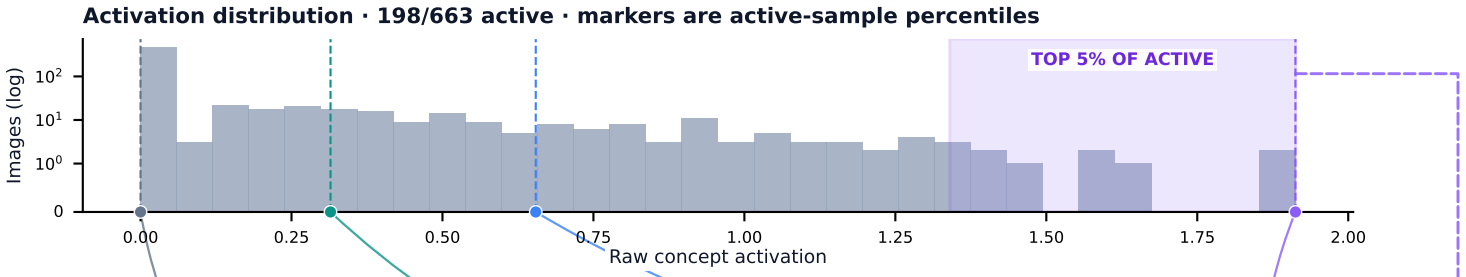
Literature source: Williamson et al., Factors associated with COVID-19-related death using OpenSAFELY, Nature 2020

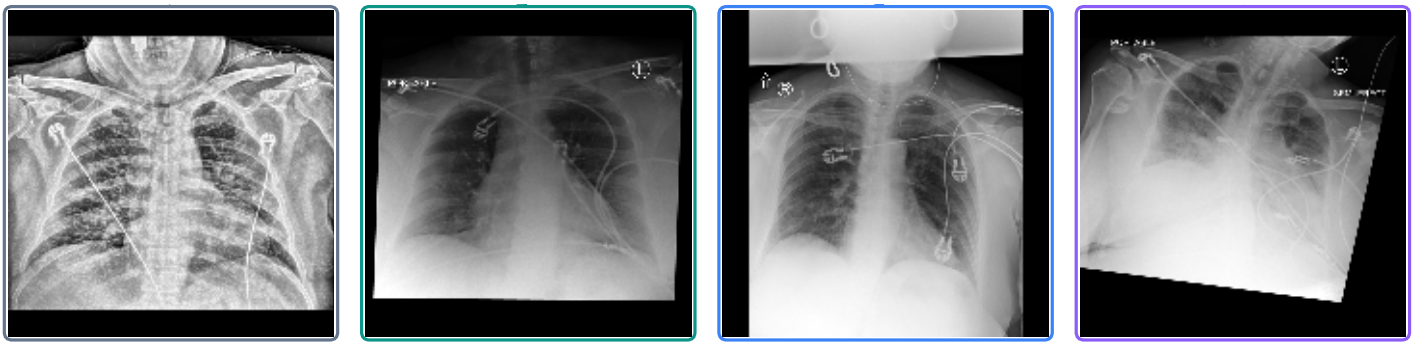
GLYPH / CONCEPT EVIDENCE

image_path concept 5744 → in_hospital_death



train: OR 1.92 (95% CI 1.61–2.30); corrected p=3e-09
test: OR 2.48 (95% CI 1.55–3.95); corrected p=0.96





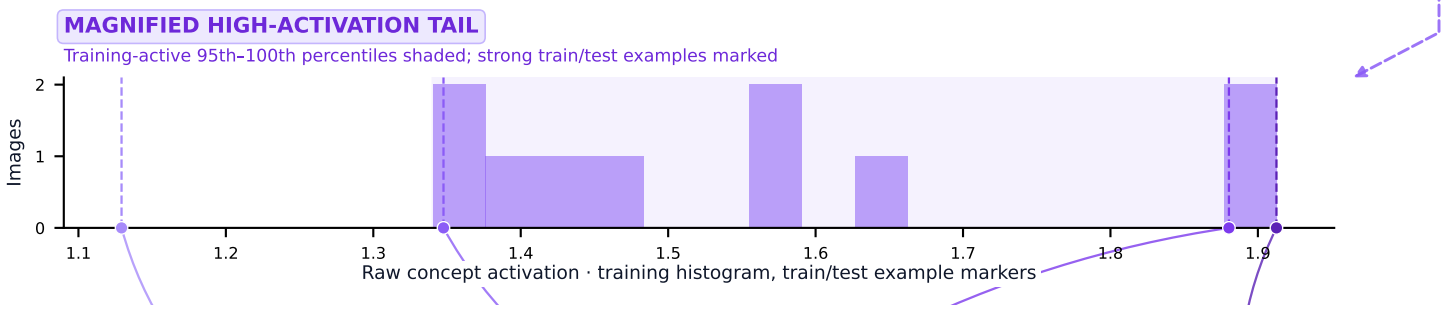
A000801 · a=0
train · discharged

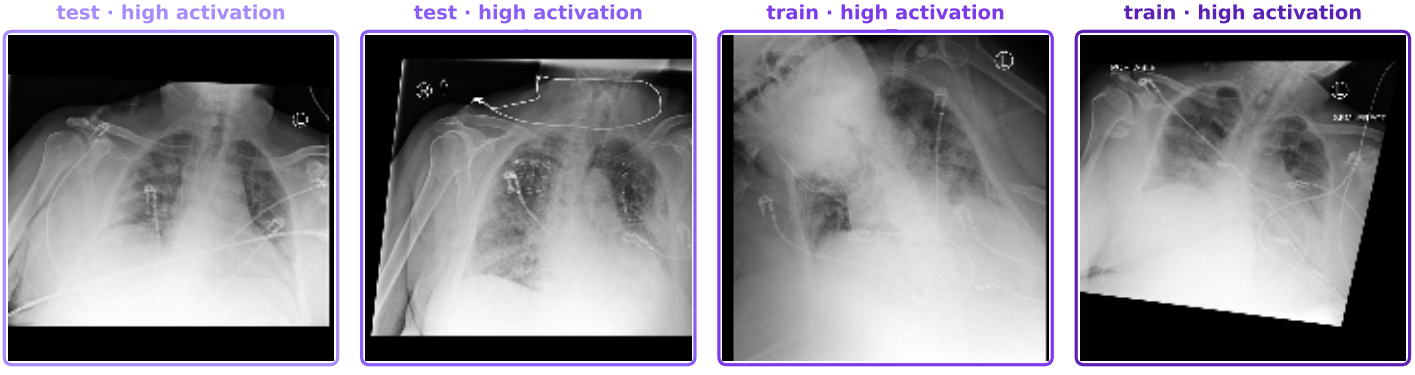
A408220 · a=0.315
train · discharged

A683530 · a=0.654
train · discharged

A632768 · a=1.91
train · later death

95–100% magnified





A712444 · a=1.13
test · discharged

A672510 · a=1.35
test · later death

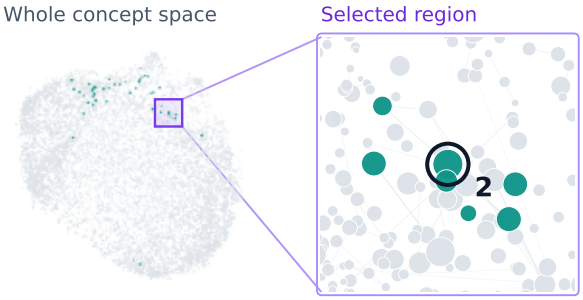
A590713 · a=1.88
train · later death

A632768 · a=1.91
train · later death

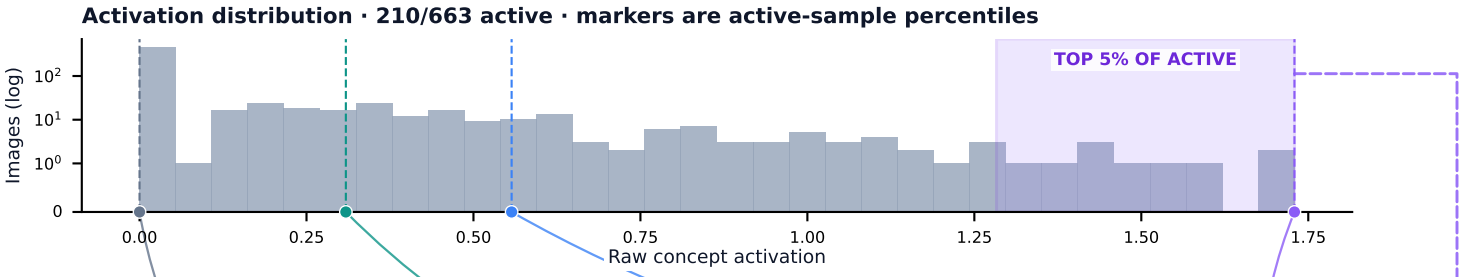
Scale images selected on training activations only. Zero is shown separately; other percentiles exclude zeros. Strong examples are selected by activation within train/test, without outcome-based selection. Images do not validate a semantic interpretation.

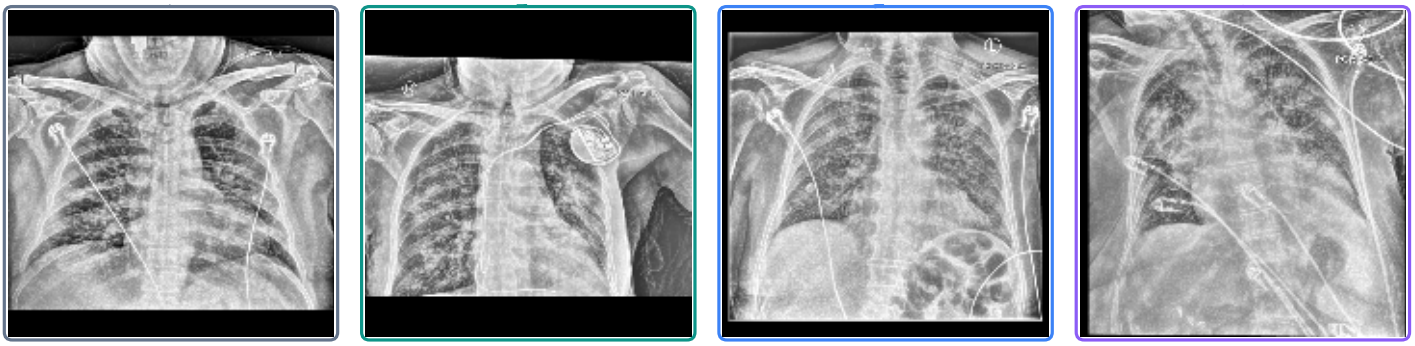
GLYPH / CONCEPT EVIDENCE

image_path concept 1027 → in_hospital_death



train: OR 1.80 (95% CI 1.51–2.14); corrected p=3.4e-07
test: OR 2.80 (95% CI 1.70–4.61); corrected p=0.36





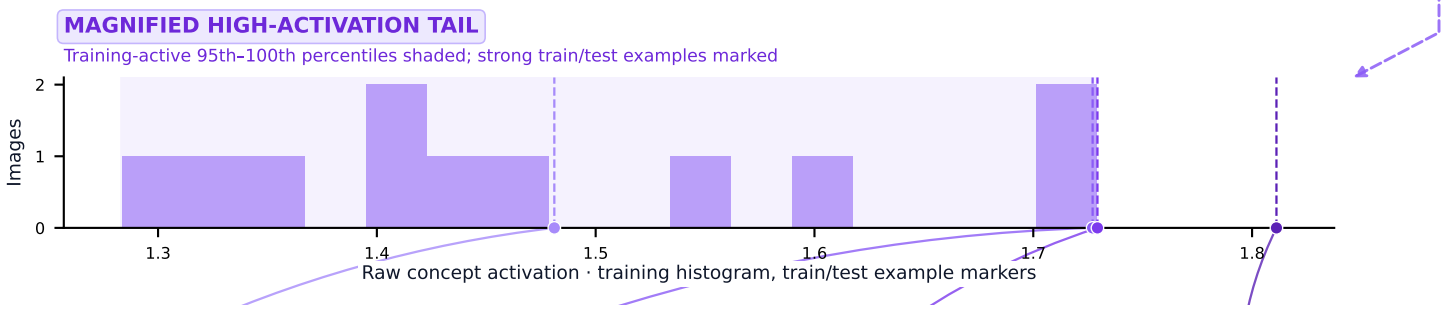
A000801 · a=0
train · discharged

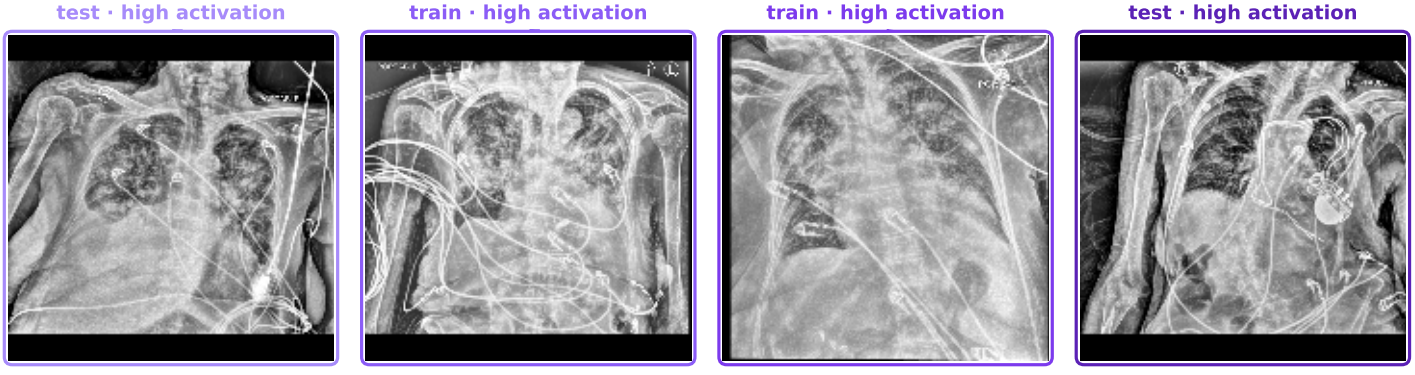
A045825 · a=0.309
train · later death

A373370 · a=0.557
train · later death

A744141 · a=1.73
train · later death

95–100% magnified





A843980 · a=1.48
test · later death

A025840 · a=1.73
train · later death

A744141 · a=1.73
train · later death

A934948 · a=1.81
test · later death

Scale images selected on training activations only. Zero is shown separately; other percentiles exclude zeros. Strong examples are selected by activation within train/test, without outcome-based selection. Images do not validate a semantic interpretation.